



香港城市大學
City University of Hong Kong
Innovating into the Future

CS5495 Explainable AI

Lecture 1 - Introduction

2026/27 Semester A

Prof. Antoni B. Chan

Course General Info

- Teaching Team

- Prof. Antoni B. CHAN

- abchan@cityu.edu.hk, Office: AC1-G7311

- TAs:

Name	email	Name	email
Guoliang LIN	guolialin2-c@my.cityu.edu.hk	Xin HUANG	xhuang899-c@my.cityu.edu.hk
Siyuan LIU	-	Feiyu CHEN	feiyuchen3-c@my.cityu.edu.hk
Bo FANG	bofang6-c@my.cityu.edu.hk	Shunkai YU	

- Office hours: see Canvas, under Syllabus.

- Canvas-based course site

- It is your own responsibility to check Canvas and University e-mail account regularly for announcements and updates.

Class Times

- Lecture
 - Wedn, 19:00-20:50 (LI 2505)
- Tutorials
 - Wedn, 21:00-21:50 (LI 4412) – Antoni Chan, Guoliang Lin
 - Wedn, 21:00-21:50 (LI 4307) – Xin Huang
 - (I will float between rooms)

Teaching Activities

- **Lectures** (2 hours per week)
 - Engage with selected XAI methods, and the intuition and principles behind them.
 - Illustrated with both toy and real-world examples.
- **Tutorials** (1 hour per week)
 - use XAI methods on small examples to gain better understanding.
- **Assignments** (2)
 - Implement and apply XAI methods to small AI models, interpret the results.
 - Observe the effectiveness of the methods, and evaluate their differences.
 - Assignment 2 is a Team-Based Learning (TBL) activity (groups of 4).
- **Course Project**
 - Implement and apply XAI methods to analyze real-world AI models.
 - Course report
 - Presentation: short-video & Q/A.
 - 2 students per group.

Assessment

- **Coursework (70%)**
 - Tutorial exercises (10%) – due 2 weeks after lecture.
 - Assignments (40%)
 - Course Project (30%) – due Week 13
 - Project proposal, report, and presentation.
- **Final Exam (30%)**
- **Note:** Must get at least 30% on final exam and 30% on course project to pass the course.

GenAI Usage

- **Policy:**

- For tutorial and assignments, **can** use GenAI.
- For course project, Students can **only** use GenAI for **editing** the English of the report, **debugging** code, or **brainstorming**. GenAI cannot be used for other aspects, e.g., writing code, running experiments, or analyzing experiment results.

- **GenAI resources:**

- CityUHK portal: <https://www.cityu.edu.hk/portal/category/5>
 - general chatbot
- GPT-Enhanced Learning (GEL): <https://gel-student.cs.cityu.edu.hk>
 - Specific for course

Programming

- **Python**
 - high-level scripting language
 - Jupyter, aka iPython Notebooks (ipynb)
 - interactive computational environment in web browser.
 - can combine text (Markdown) with Python code and output.
 - Libraries
 - numpy – arrays, linear algebra
 - matplotlib, pylab – plotting
 - scikit, scikit-learn – scientific computing, machine learning
 - pytorch – deep learning
 - shap, lime, captum, eli5 – XAI toolkits
 - Introduction later today

Assignments

- **Assignment/Projects**

- Code/Report submission
 - Jupyter notebooks (ipynb)
 - Python scripts (for project, if necessary)

- **Team-based Learning (TBL)**

- work in groups on Assignment 2.
- make a few memes about your results/analysis.

Course Abstract

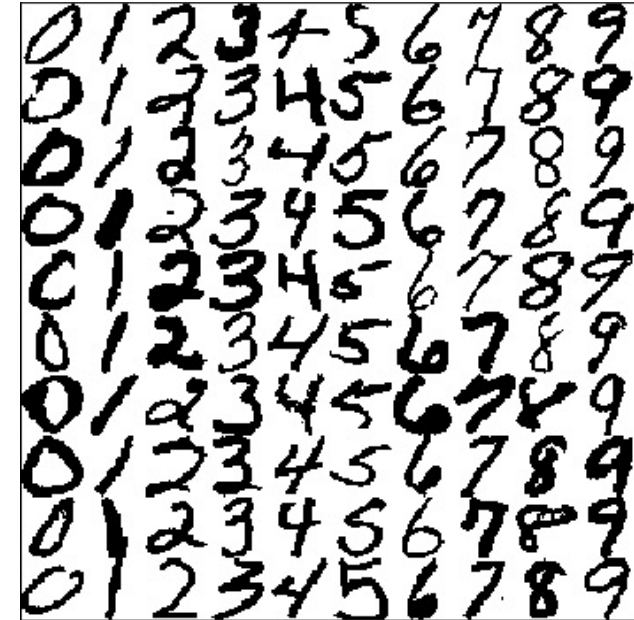
- The goal of this course is to introduce students to explainable AI (XAI) methods, which aim to explain the predictions of black-box AI models.
 - Such explanations are important for establishing good communication, trust, clarity, and understanding of AI models, which can increase their adoption in critical systems and for solving complex problems.
- This course is intended to give a broad overview of different XAI methods from a practical standpoint, with a focus on applying XAI and interpreting and analyzing the AI systems.
- At the end of the course, students will have both working knowledge of and practical experience implementing and applying XAI on different AI models and different domains.

CILOs

1. Identify and explain common explainable AI methods.
2. Implement explainable AI methods.
3. Apply explainable AI methods to analyse real-world AI models.
4. Evaluate the effectiveness of different explainable AI methods and discuss their advantages and disadvantages.

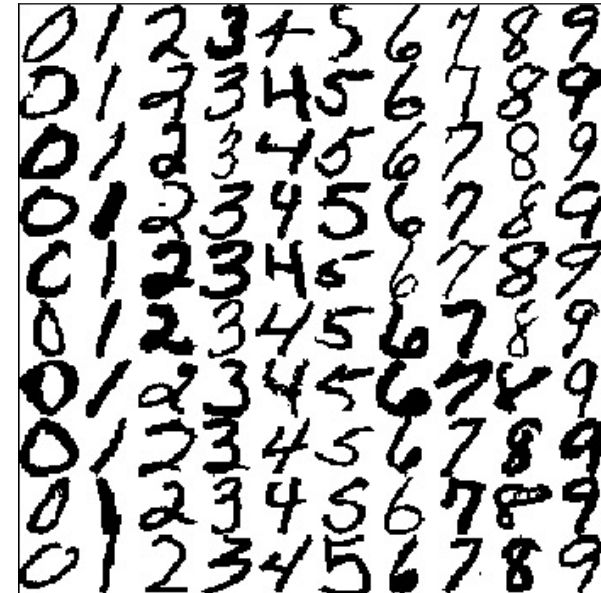
What is Machine Learning?

- Arthur Samuel, 1959
 - *Machine Learning*: field of study that gives computers the ability to learn without being explicitly programmed.
 - e.g. computer learns to play checkers by playing against itself.
- There are many applications that are difficult to program by hand.
 - Example: Recognizing handwritten digits in an image.



What is Machine Learning?

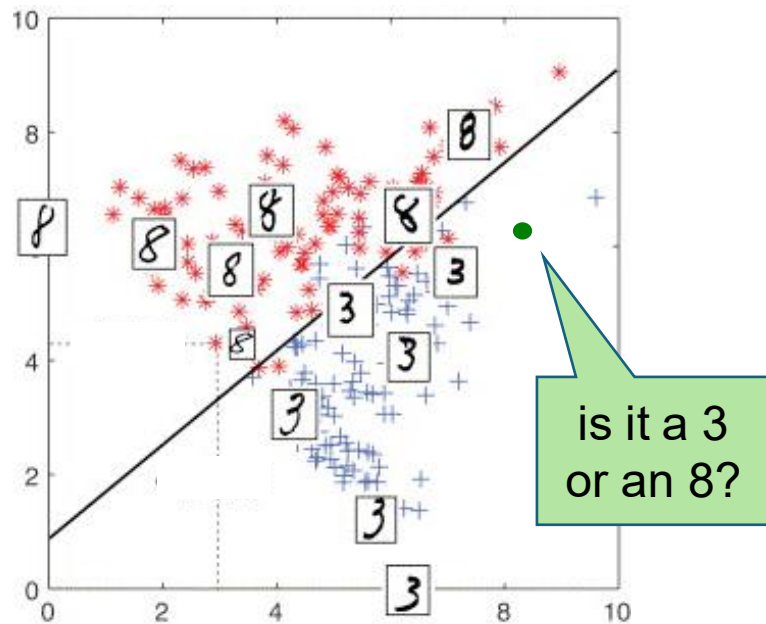
- Example: Recognizing handwritten digits in an image
 - 28x28 image $\rightarrow$ 784-dim vector
 - a lot of variations & permutations
 - difficult to identify rules & code by hand
- ML solution:
 - gather some example data.
 - train computer to discover differences automatically



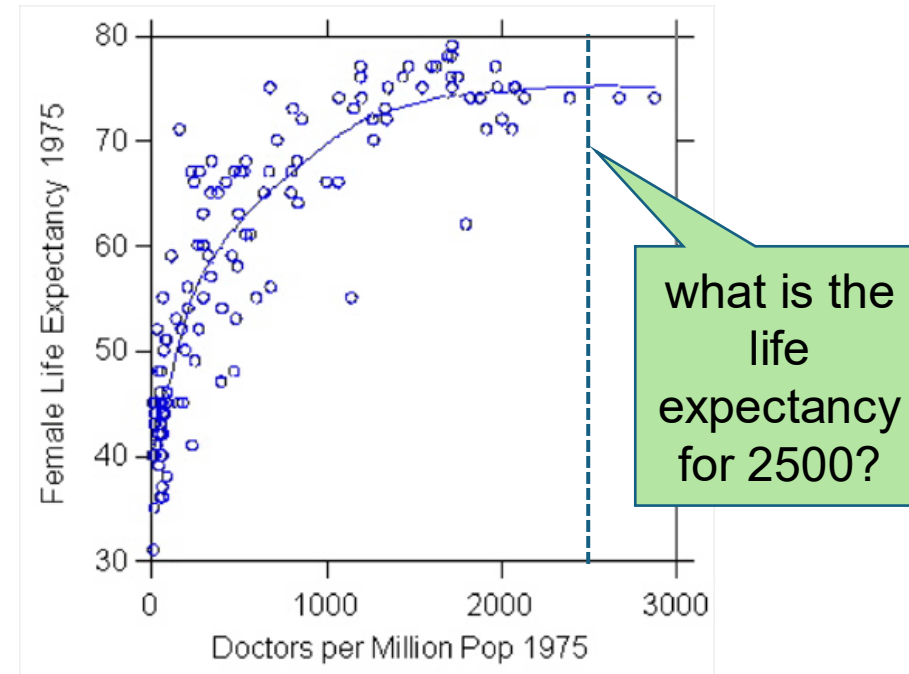
Supervised Learning

- Training data has inputs and outputs
 - e.g., digit recognition (input=image, output=digit)
 - learn a function mapping inputs to outputs

Classification

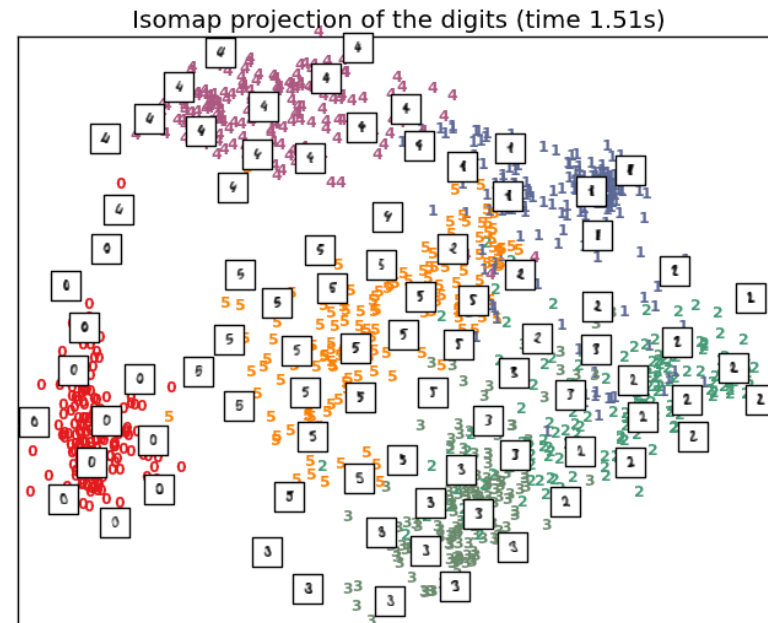
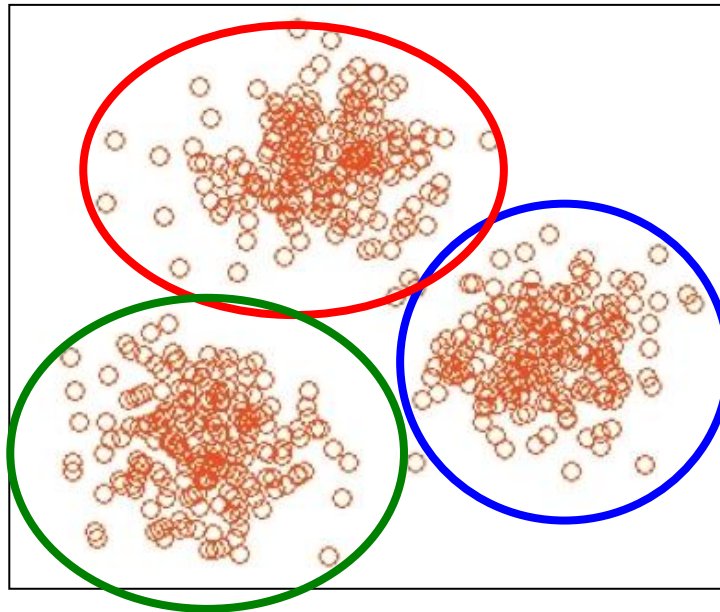


Regression



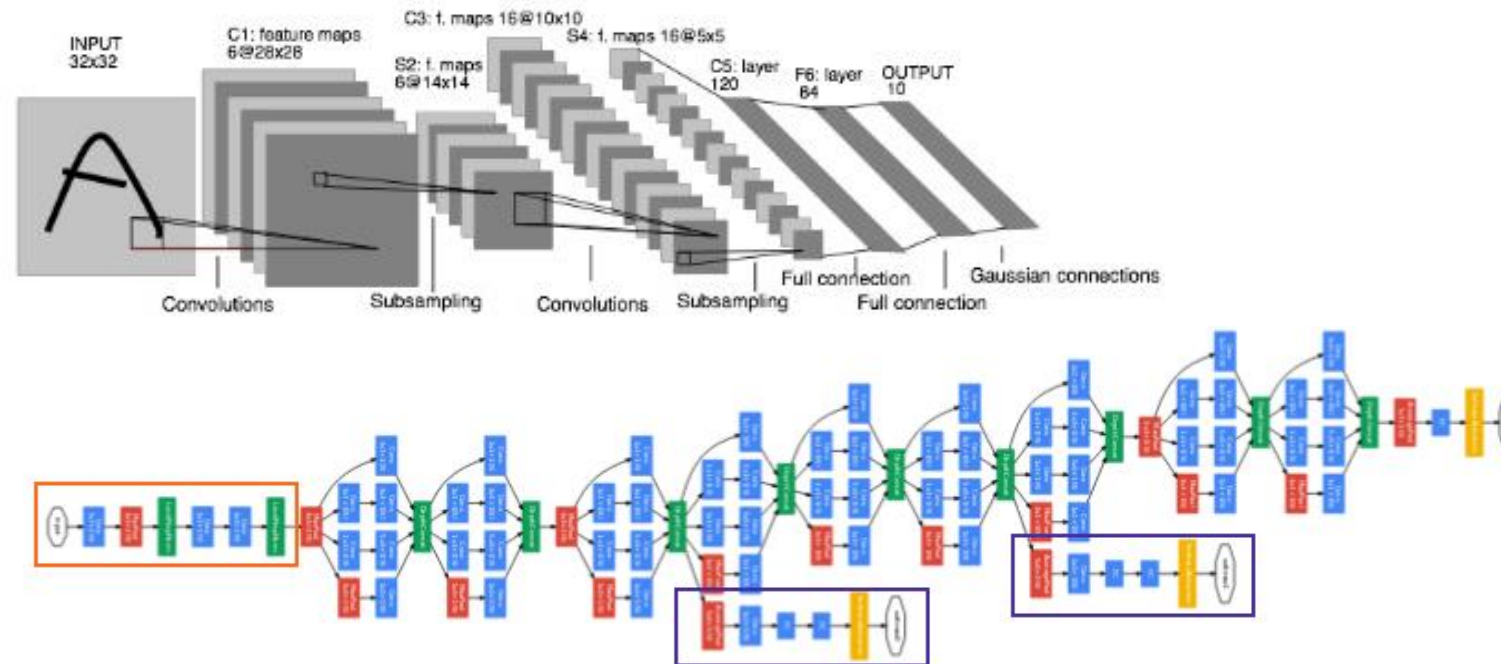
Unsupervised Learning

- Training data only has inputs (no outputs)
 - e.g., collection of web documents
 - clustering - discover groups of similar examples.
 - visualization - project high-dim data to 2 or 3-dimensions.



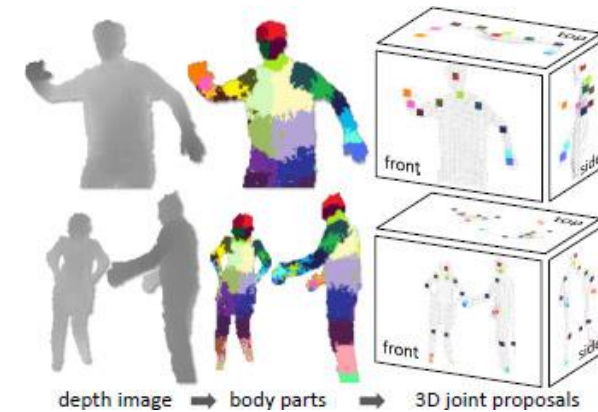
Deep Learning

- Deep learning is supervised/unsupervised learning using multi-layer neural networks.
 - Improved training algorithms to prevent overfitting
 - Faster parallelization (GPUs)
 - More data (millions of examples)

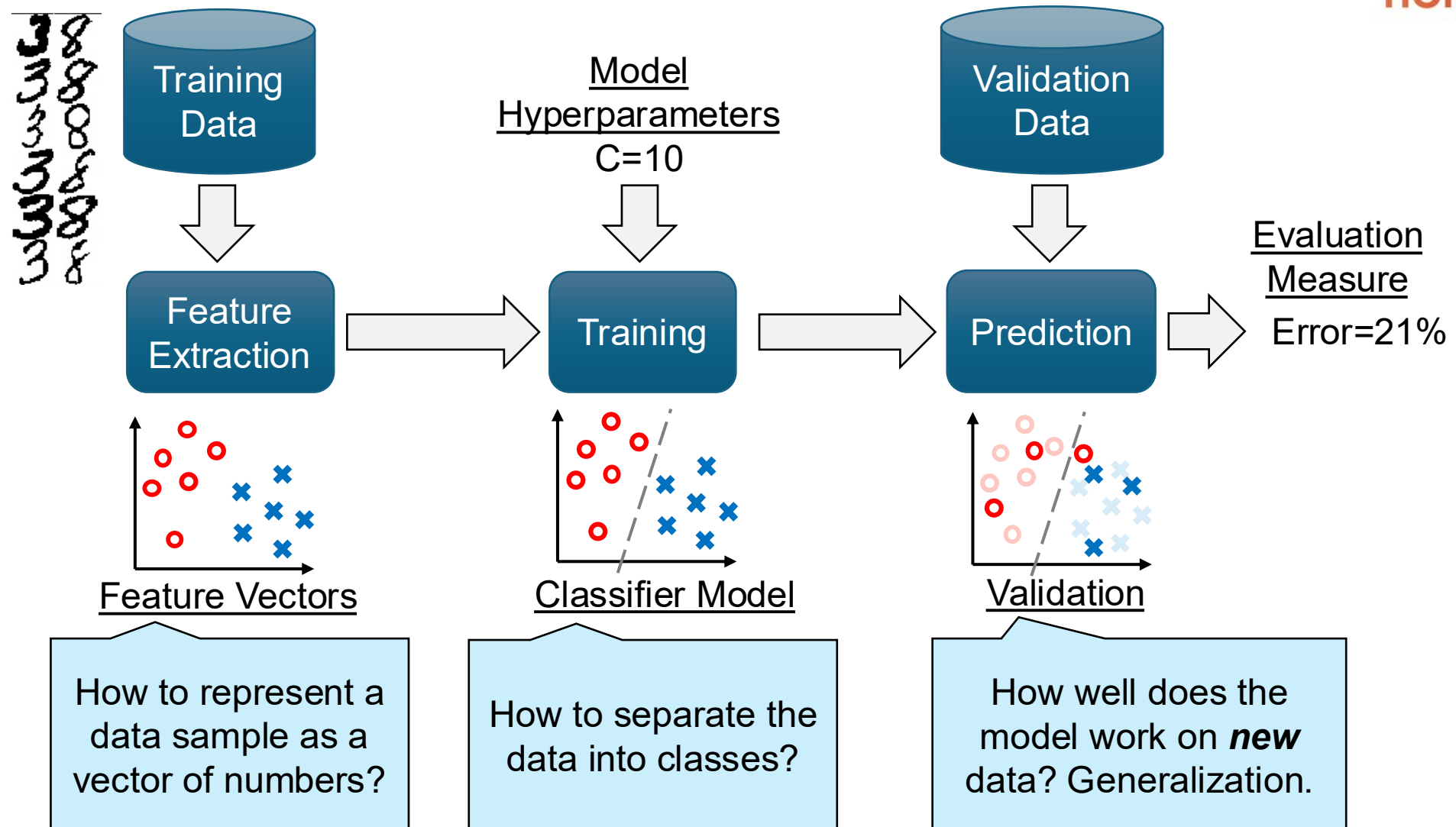


ML in the Real World

- Google
 - spam email classifier, speech recognition, AlphaGo, machine translation, image search, quick links,
- Face detection & recognition
 - digital cameras, Google street view, Facebook
- Business
 - credit card fraud detection
 - stock trading (portfolio optimization)
- Recommendation systems
 - Netflix, Amazon
- Human pose recognition (Kinect)
- Controllers (reinforcement learning)
- Self-driving Cars
- Generative AI (ChatGPT, Dall-E)

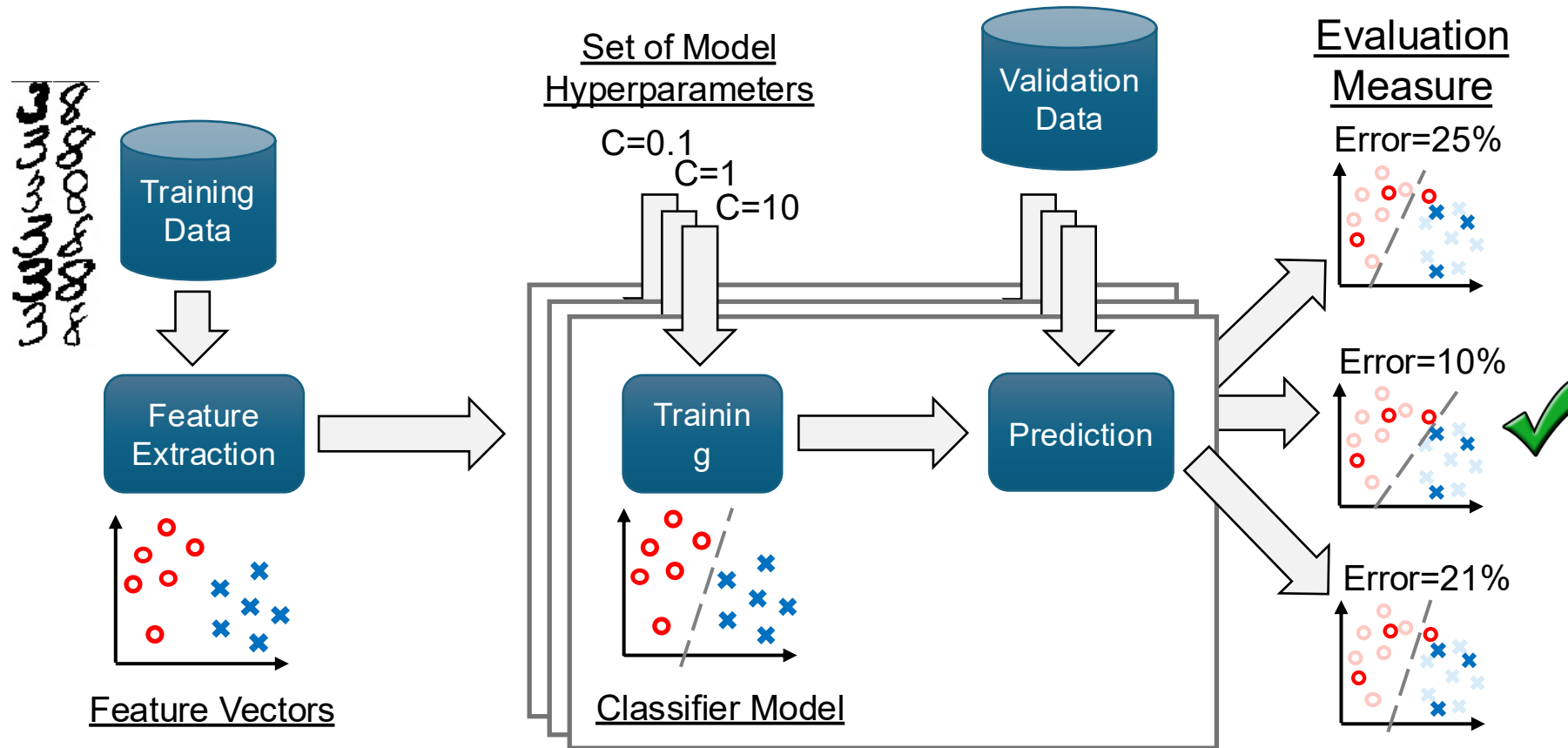


ML Training Pipeline



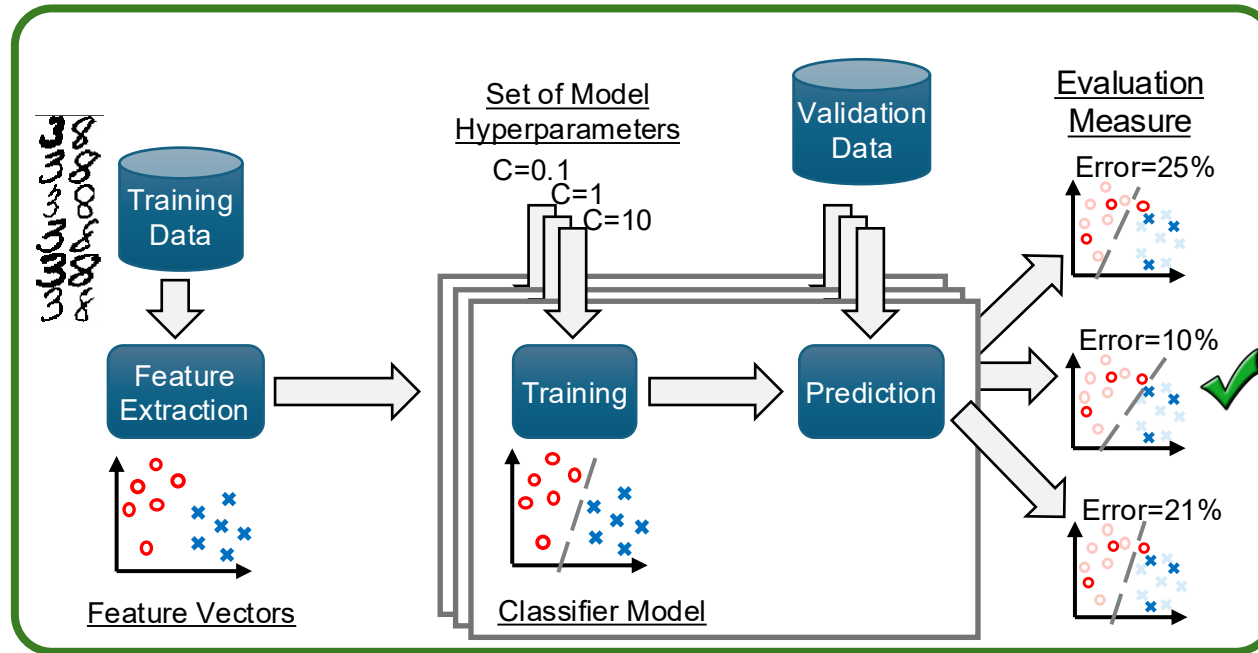
Model Selection Pipeline

- Select hyperparameters that yield low generalization error.



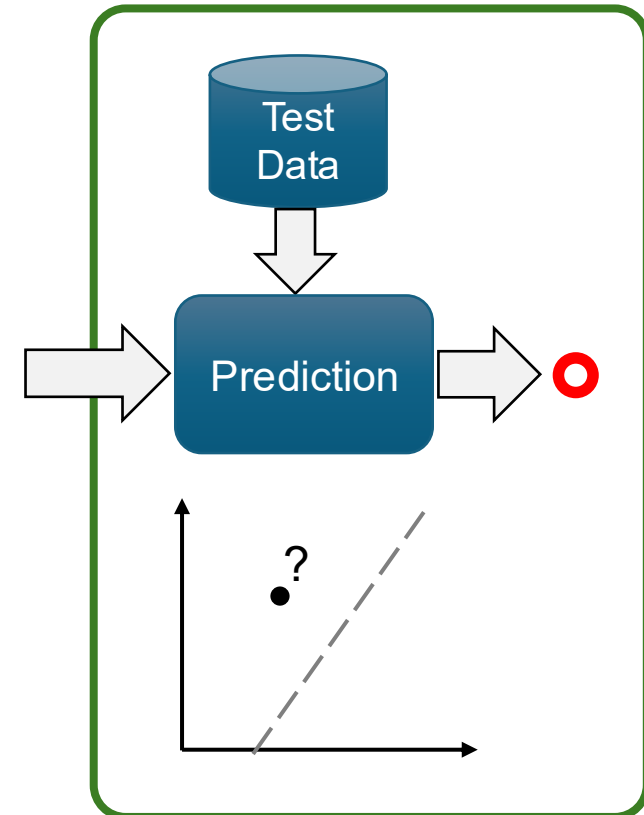
Training & Inference Stages

Training Stage



Typically slow, because need to try different hyperparameters and iterative over a training algorithm.

Testing (Inference) Stage



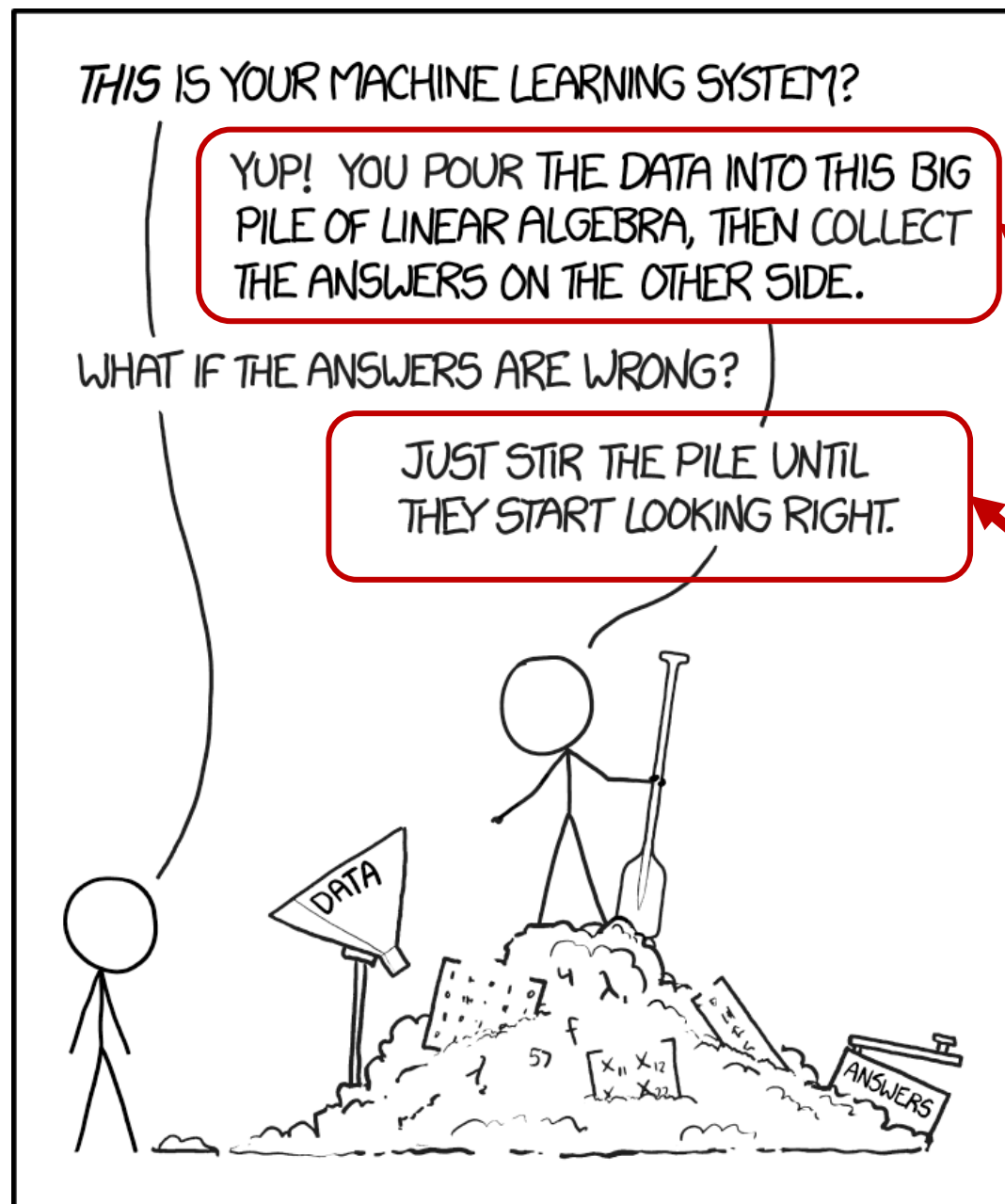
Typically fast, because the model is already estimated.

Building Blocks for ML

- What are the tools needed for Machine Learning?
 - *linear algebra*
 - matrices, inverse, eigenvector, SVD, ...
 - *probability & statistics*
 - random variables, expectation, Bayes' theorem, ...
 - *optimization algorithms*
- ML is built on solid foundations of mathematics.
 - can build and train complex models
- ML premise:
 - “difficult to identify rules & code by hand”
 - “train computer to discover differences automatically”

What exactly did
the model learn?





We only give ML the input and output pairs for training.

No idea how it works inside (black box).

Motivation

- **Can't blindly trust an AI model!**
 - A model trained to classify a tank as American or Russian used the lighting conditions of the image.
 - A machine learning model trained to detect whales learned to rely on artifacts in audio files instead of basing the classification on the audio content ([DeLMA and Cukierski 2013](#)).
 - An image classifier learned to use text on images instead of visual features ([Lapuschkin et al. 2019](#)).
 - A wolf versus dog classifier relied on snow in the background instead of image regions that showed the animals ([Ribeiro, Singh, and Guestrin 2016](#)).
- **Need a way to debug the AI model...**

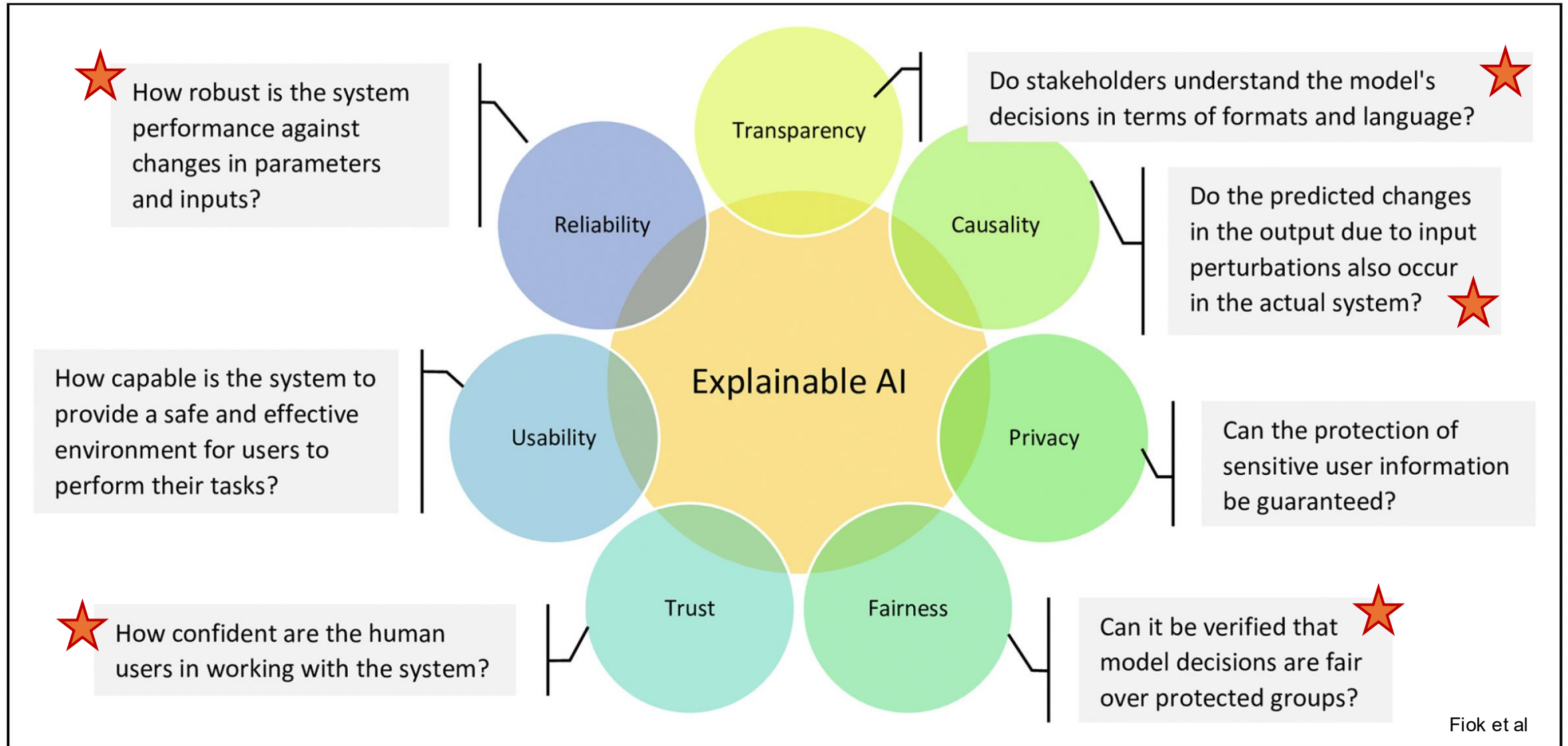
People with no idea
about AI, telling me my
AI will destroy the world

Me wondering why my
neural network is
classifying a cat as a dog..

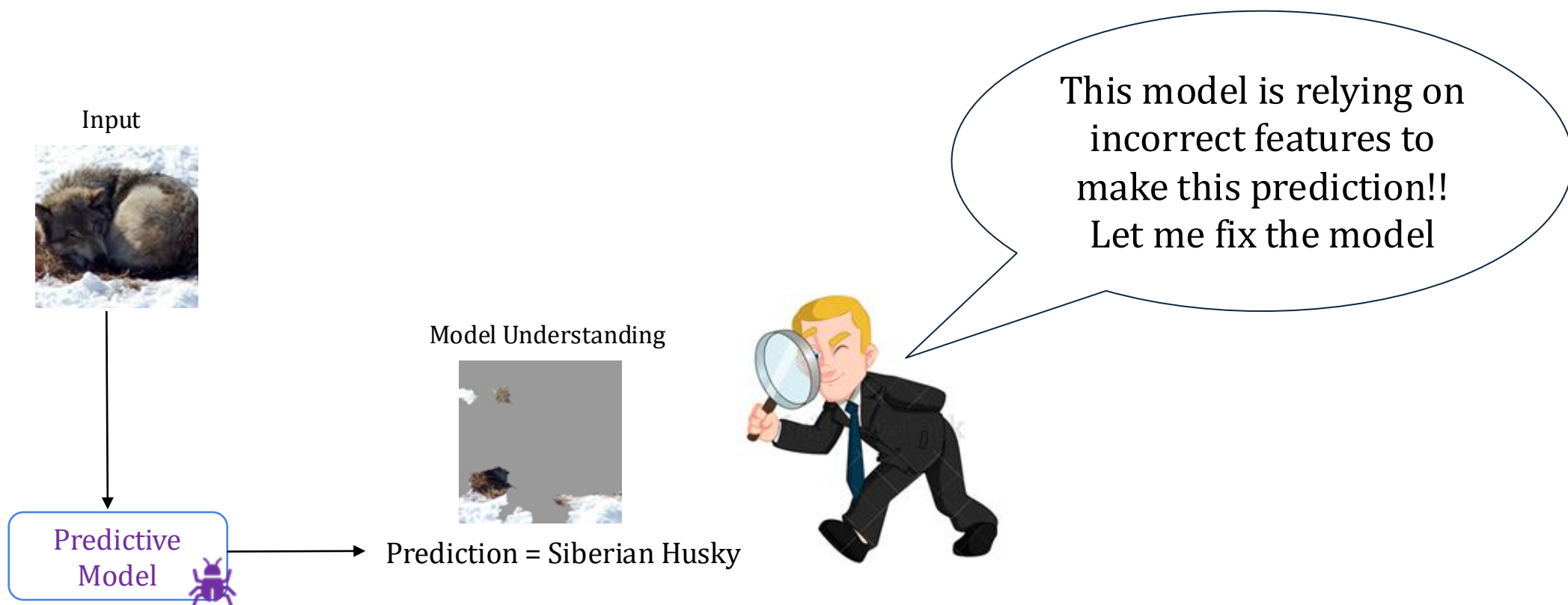


What is XAI?

A major component of XAI is understanding **why** the model made the prediction.

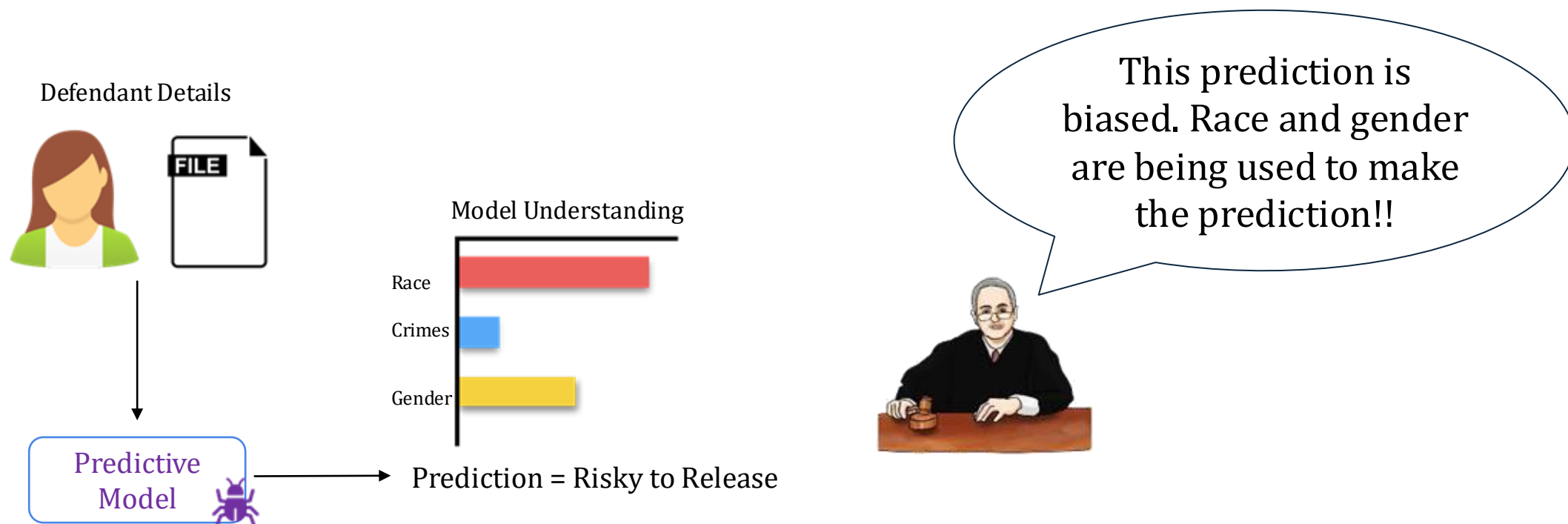


Why Model Understanding?



Model understanding facilitates debugging.

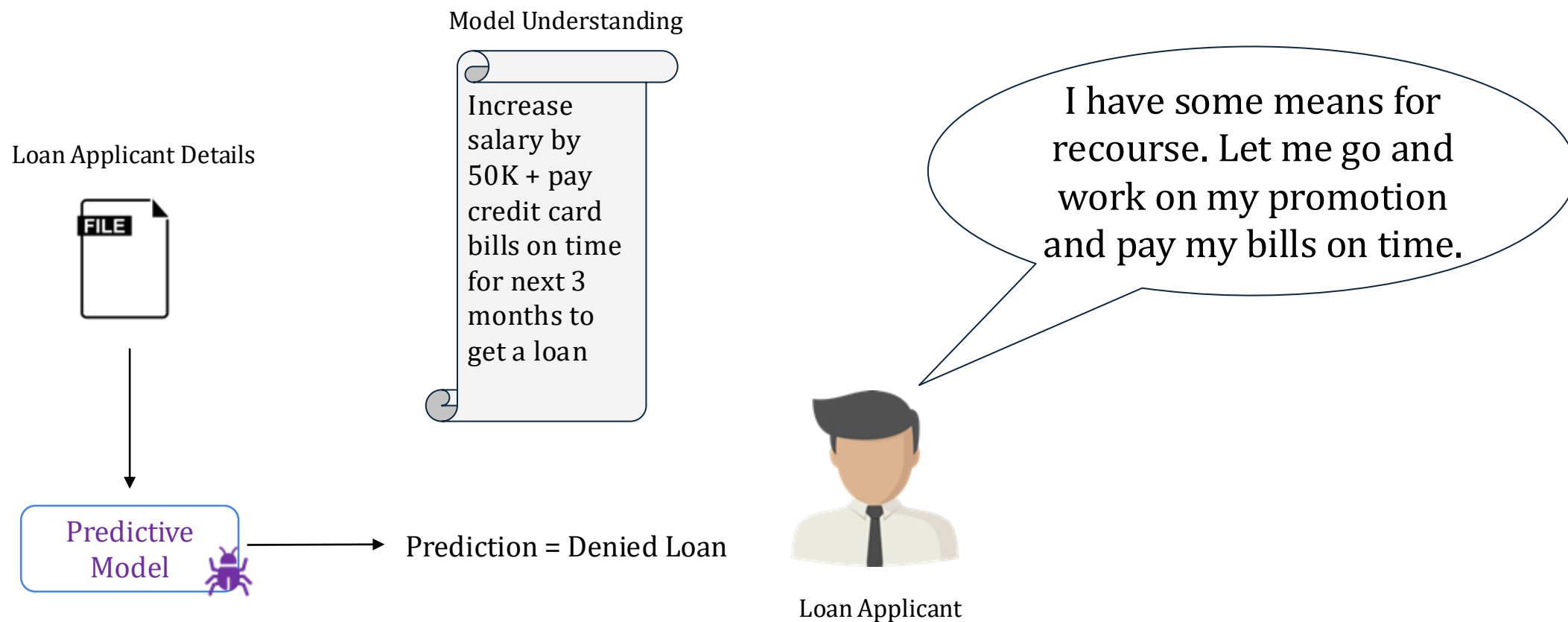
Why Model Understanding?



Model understanding facilitates bias detection.

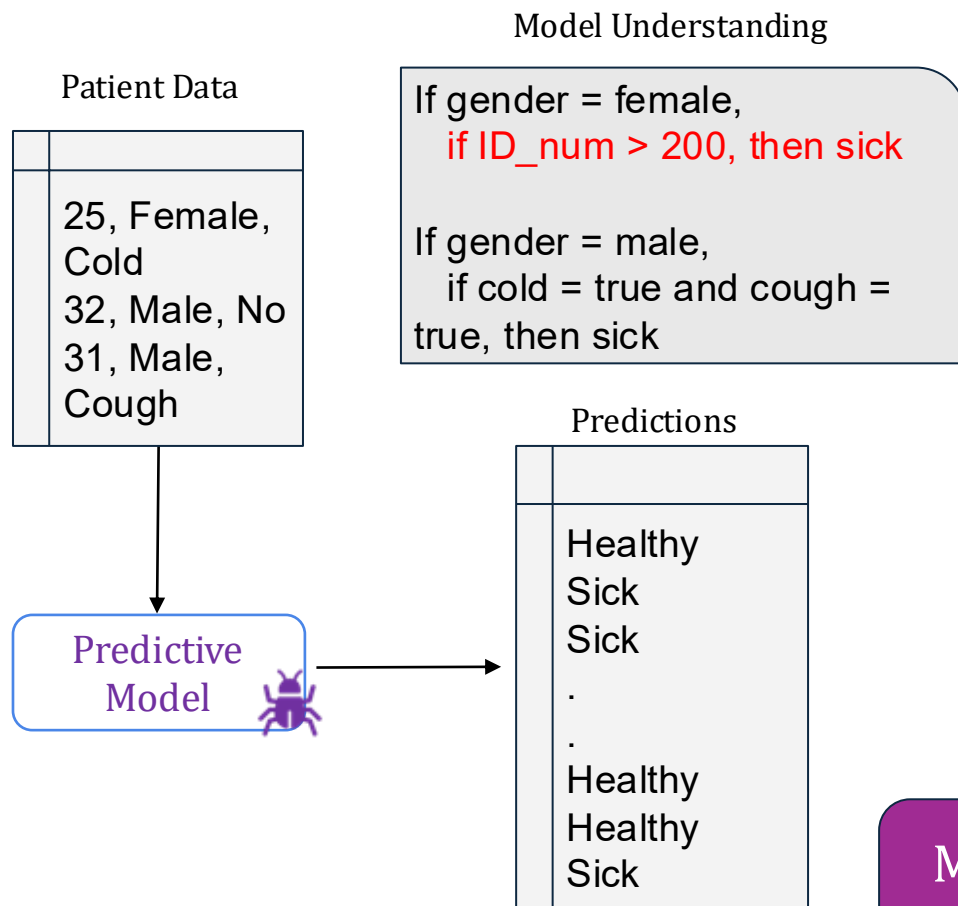
[Larson et. al. 2016]

Why Model Understanding?



Model understanding helps provide recourse to individuals who are adversely affected by model predictions.

Why Model Understanding?



This model is using irrelevant features when predicting on female subpopulation. I should not trust its predictions for that group.



AUTHORITY

This cannot be approved!

Model understanding helps assess if and when to trust model predictions when making decisions.

Serving Multiple Purposes

- Explanations serve different purposes depending on stakeholder.

Utility

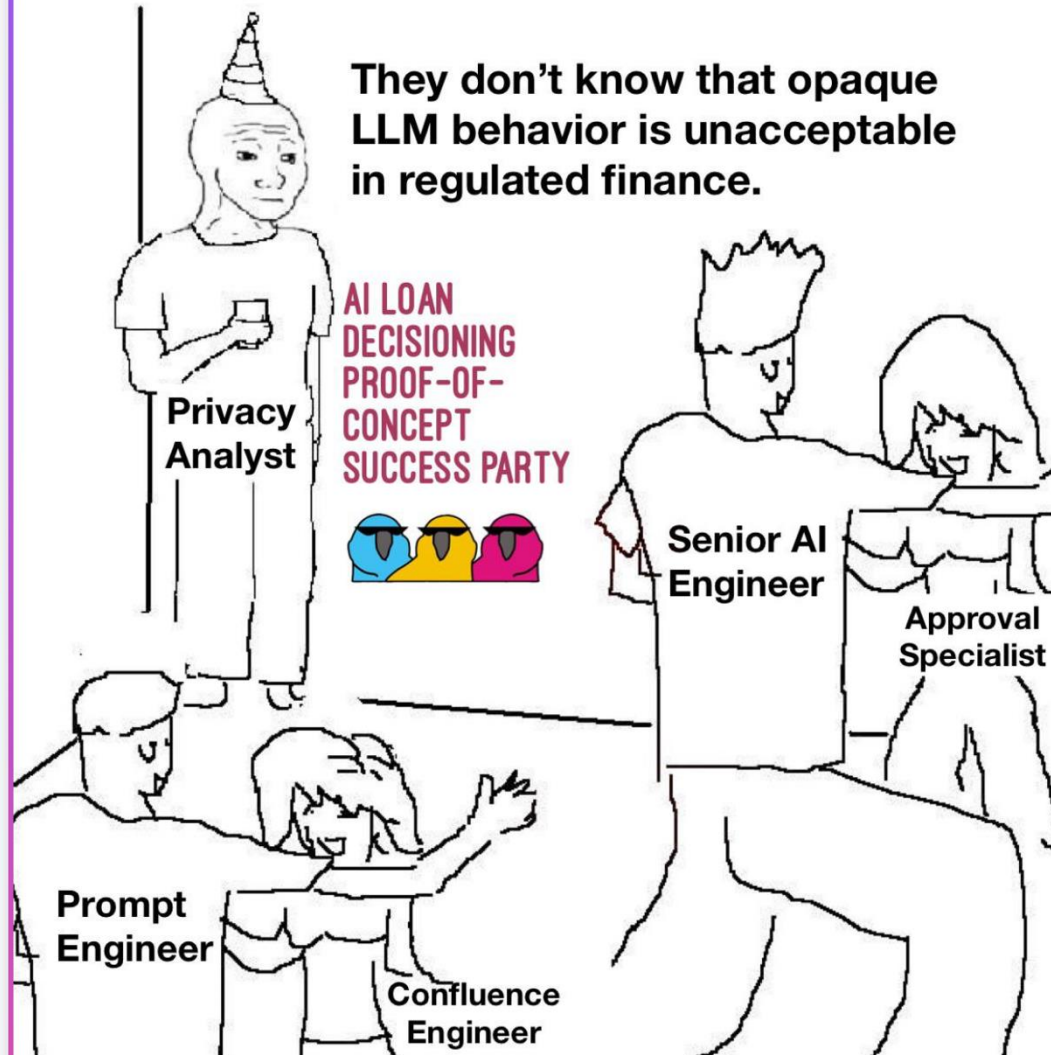
- Debug model
- Improve model
- Gain insight on the task
- Vet models to assess suitability for deployment
- If and when to trust model predictions
- \Bias Detection
- Recourse

Stakeholders

- Researchers and engineers
- Decision makers (e.g., doctors, judges)
- Regulatory agencies (e.g., FDA, European commission)
- End users (e.g., loan applicants)
- Auditors (e.g., investigators)

Proof Of Concept Utopia

#machine learning #Llm #Fintech #Proof of concept #Regulatory compliance #Black box algorithms
#Explainable ai #Financial technology



What's a Good Explanation (for Humans)?

- **Contrastive** – how does the prediction change if the input was different? (counter-factual)
- **Selected** – one to three reasons are sufficient
- **Social** – interaction between explainer and receiver. The human context matters. Who is the explanation for?
- **Focus on abnormal** – humans prefer abnormal causes (rare features) over normal causes.

What's a Good Explanation (for Humans)?

- **Truthful** – the causes are correct (fidelity, faithfulness)
- **Prior beliefs** – humans prefer explanations that conform to prior belief
- **General and probable** – the cause applies to many samples, and is not too specific.



How to Achieve Interpretability?

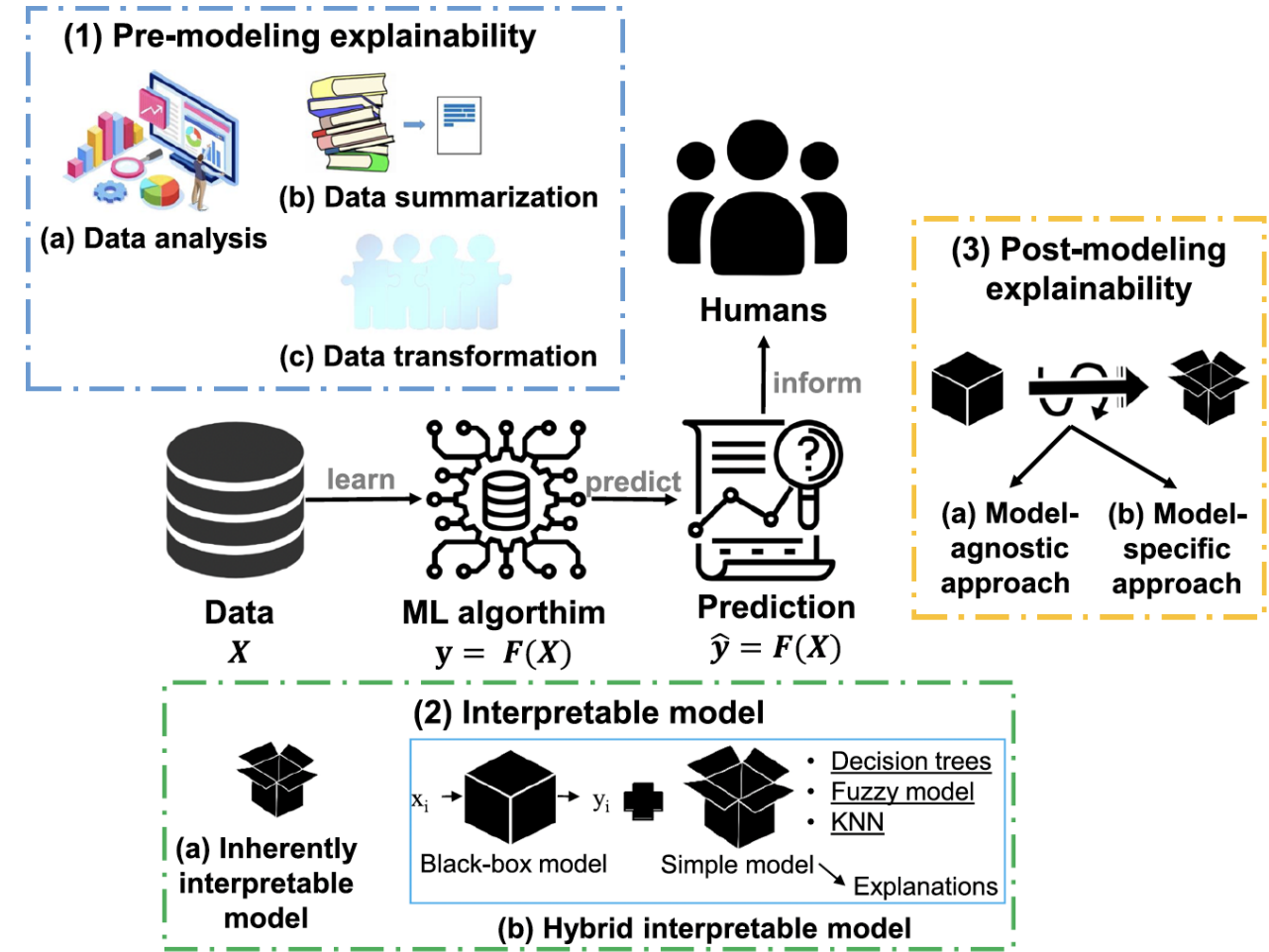
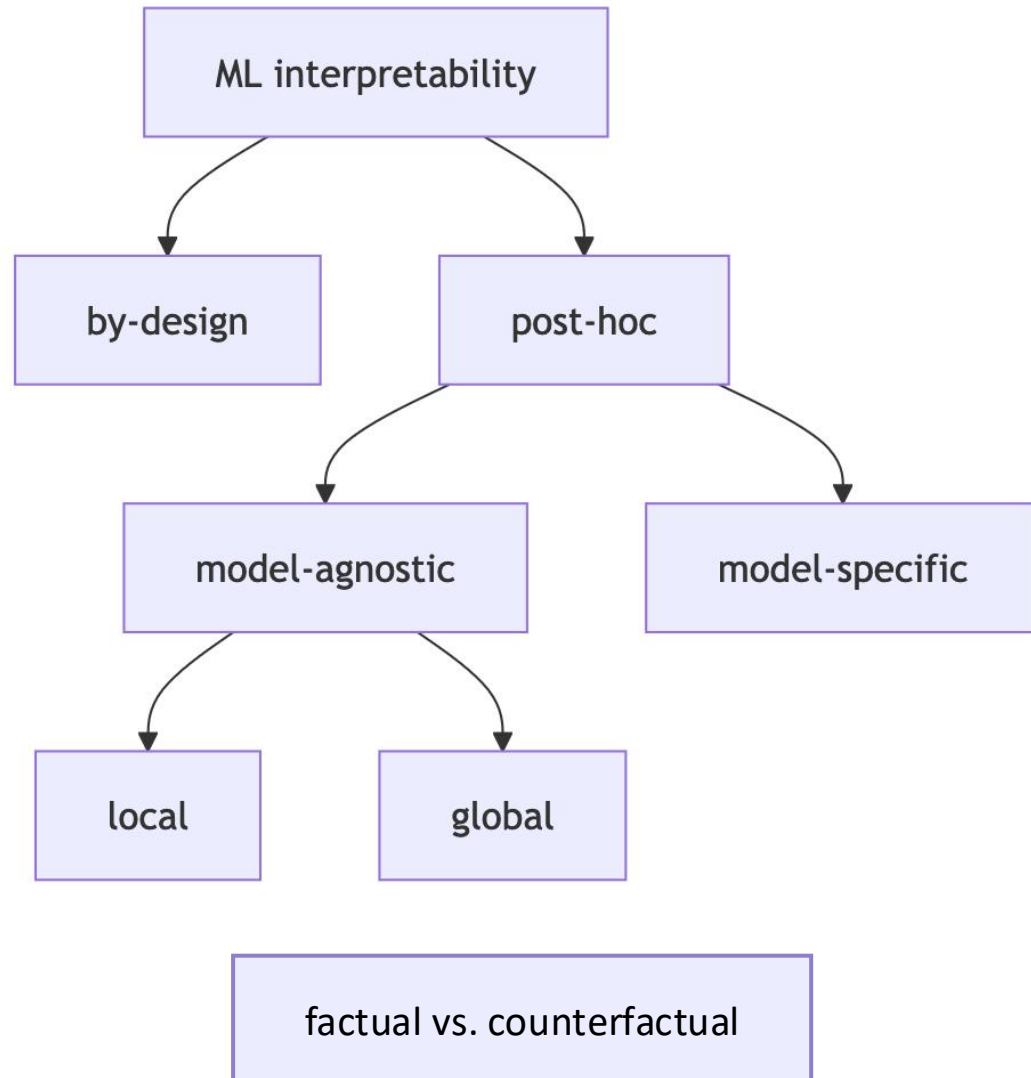
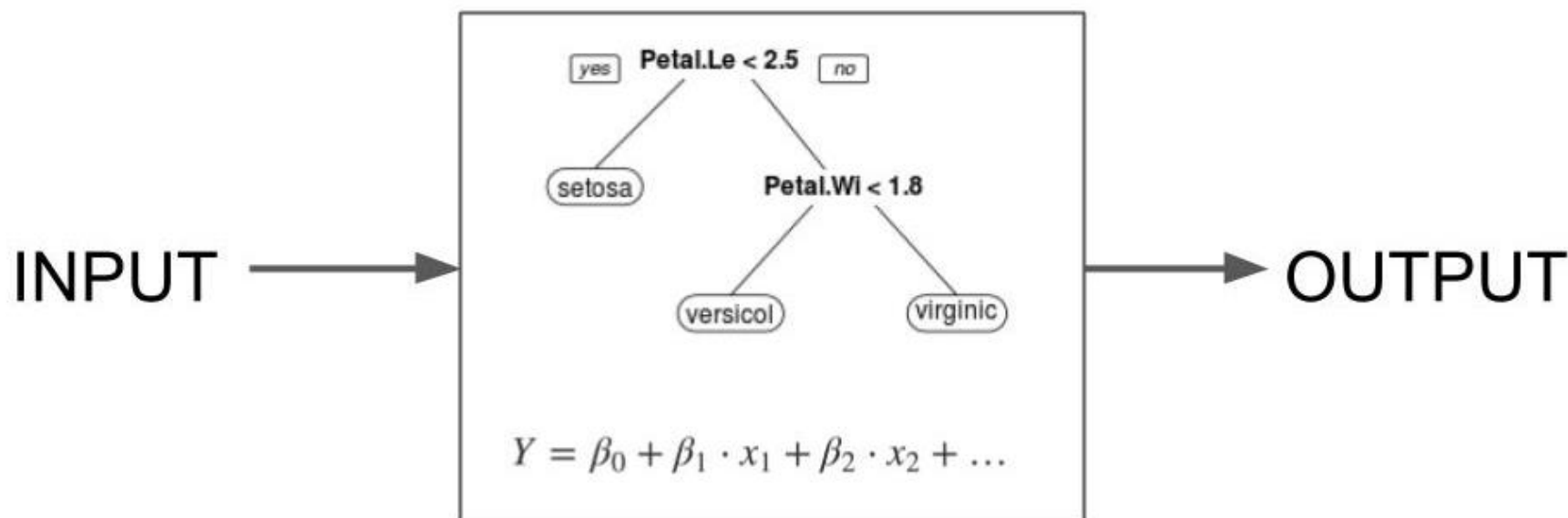


Fig. 3 Conceptual diagram that represents the three degrees of explainability, which include (1) the pre-modeling explainability, (2) the interpretable mode, and (3) the post-modeling explainability

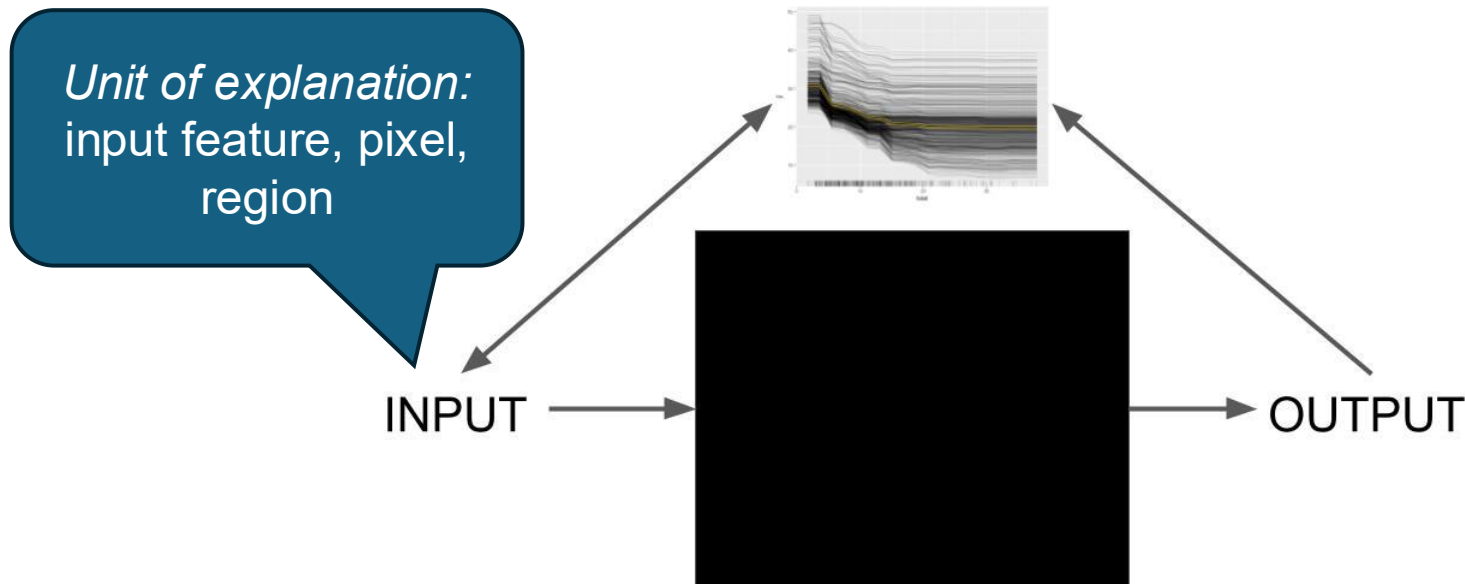
Interpretable Models

- The model is interpretable by design
 - The whole model, part of the model, or just the predictions
- E.g., linear regression, decision tree, prototype-based NN



Model-Agnostic Post-Hoc

- **Model-agnostic** – ignore the internal workings of the model (black-box)
 - **Local**: explain individual predictions
 - **Global**: explain average behavior across dataset

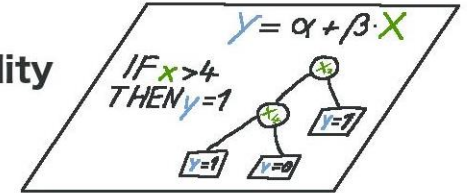


Humans



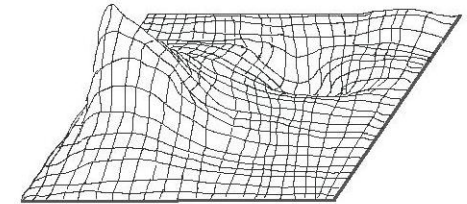
↑ inform

Interpretability
Methods



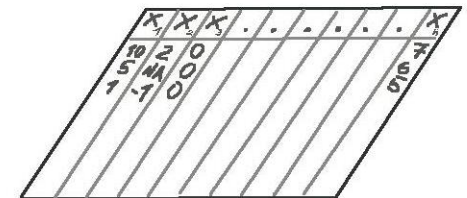
↑ extract

Black Box
Model



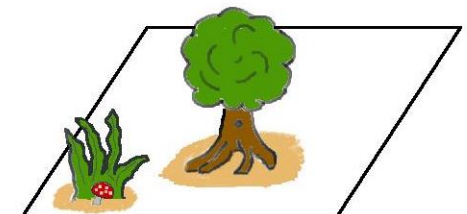
↑ learn

Data



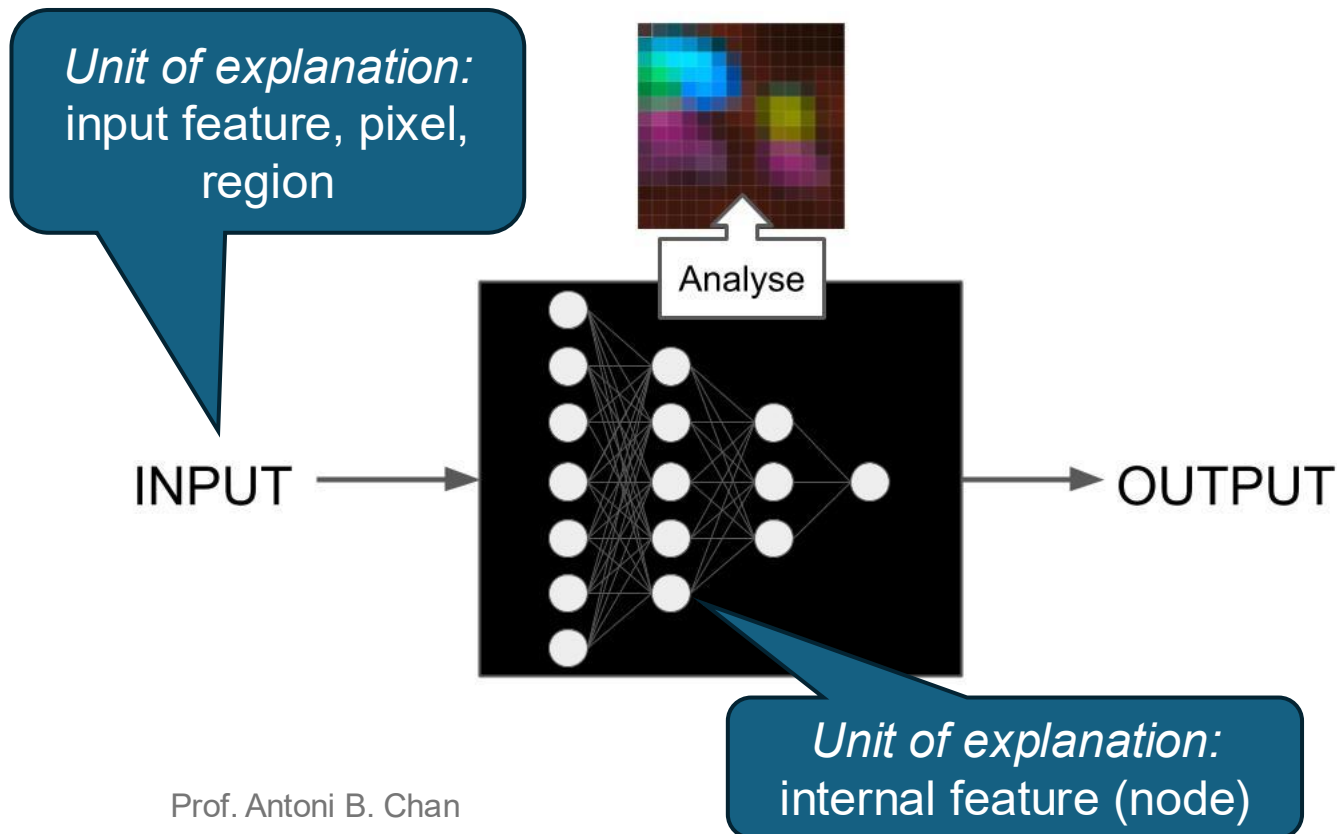
↑ capture

World



Model-Specific Post-Hoc

- **Model-specific:** interpret the inner workings of the model (white-box)
 - Need a theoretical justification why model interpretation reflects the data.



Feature visualization

CS5495 Schedule (2026/27A)

Wk	Lecture Topic	Tutorial	Assessment
1 (2/9)	Lec 1: Introduction / Python / ML <u>Traditional ML</u>	Tut 1	
2 (9/9)	Lec 2: Interpretable Linear Models	Tut 2	
3 (16/9)	Lec 3: Non-linear models, Local XAI Methods	Tut 3	
4 (23/9)	Lec 4: Global XAI Methods <u>Deep Learning</u>	Tut 4	A1 out
5 (30/9)	Lec 5: XAI for Convolutional NN	Tut 5	
6 (7/10)	Lec 6: XAI for CNNs (Part 2)	Tut 6	
7 (14/10)	Lec 7: XAI for Transformers	Tut 7	A1 due
8 (21/10)	Lec 8: Object Detection, Diffusion	Tut 8	
9 (28/10)	Team-Based Learning (TBL) – Assignment 2	-	A2 out
10 (4/11)	TBL presentations	-	A2 due, Pr. out
11 (11/11)	Lec 9: Interpretable DNNs, Theories of XAI	-	
12 (18/11)	Lec 10: LLMs & Mechanistic Interpretability (TBC)	-	
13 (25/11)	Project presentations	-	Pr due

References

- **XAI References**

- Christoph Molnar, Interpretable Machine Learning: A Guide For Making Black Box Models Explainable (2nd ed.), 2022. <https://christophm.github.io/interpretable-ml-book/>
- Explainable AI: From Simple Predictors to Complex Generative Models, Harvard. <https://interpretable-ml-class.github.io>
- Selected academic papers

- **ML References**

- Muller & Guido, "Introduction to Machine Learning with Python", O'Reilly, 2017.
- Harrington, "Machine Learning in Action", Manning Publications Co., 2012.
- C.M. Bishop, "Pattern Recognition and Machine Learning", Springer, 2006.
- Goodfellow, Bengio, Courville, "Deep Learning", MIT Press 2016. <http://www.deeplearningbook.org>
- A. Rajaraman, and J. Ullman, "Mining of Massive Datasets", Cambridge University Press, 2011. (<http://infolab.stanford.edu/~ullman/mmds.html>)
- H. Daume III, "A course in Machine Learning", (<http://ciml.info/>)

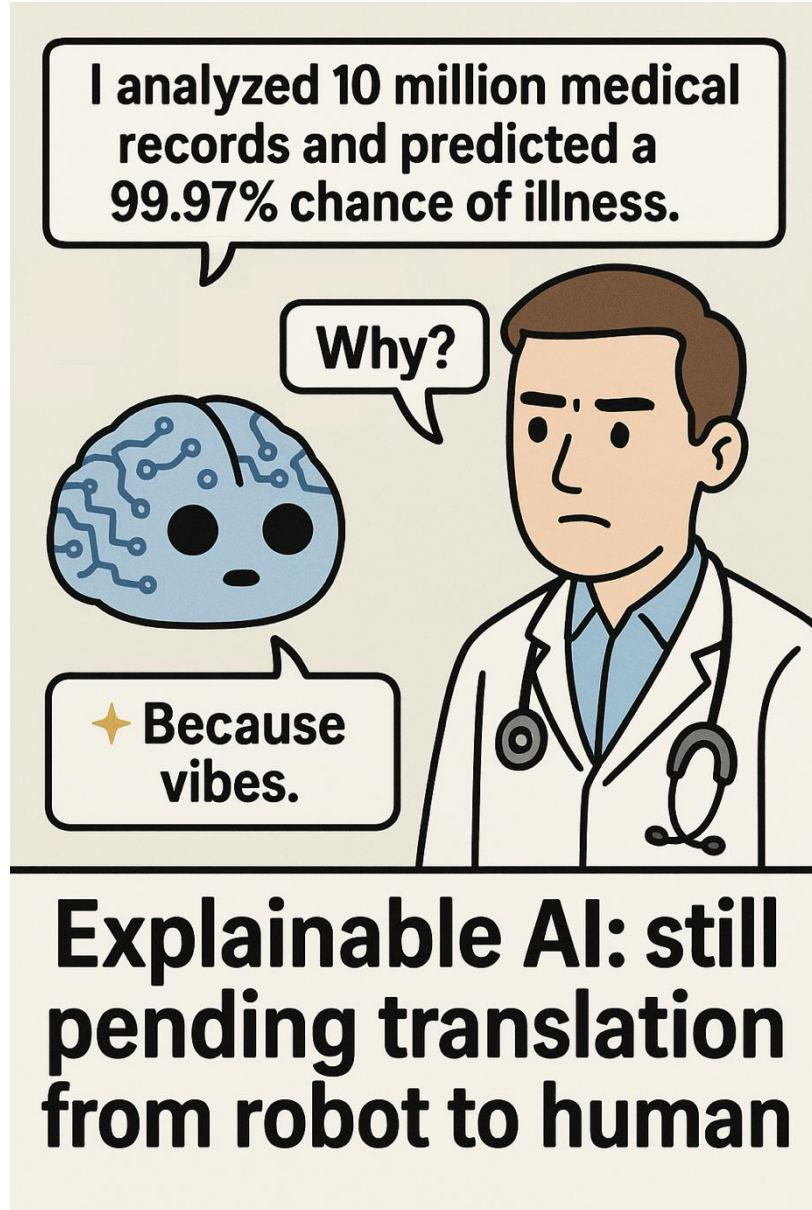
Computing Resources

- Google Colab
 - Use your gapps account: **your-EID@gapps.cityu.edu.hk**
 - Free version with computing limits.
 - Enterprise version with computing credits – we are setting this up now.
- CS Lab JupyterHub
 - Provides Jupyter notebooks on a central server for multiple users.
 - Select between CPU and GPU resource
 - dedicated for this course.
- Check Canvas page for more details.

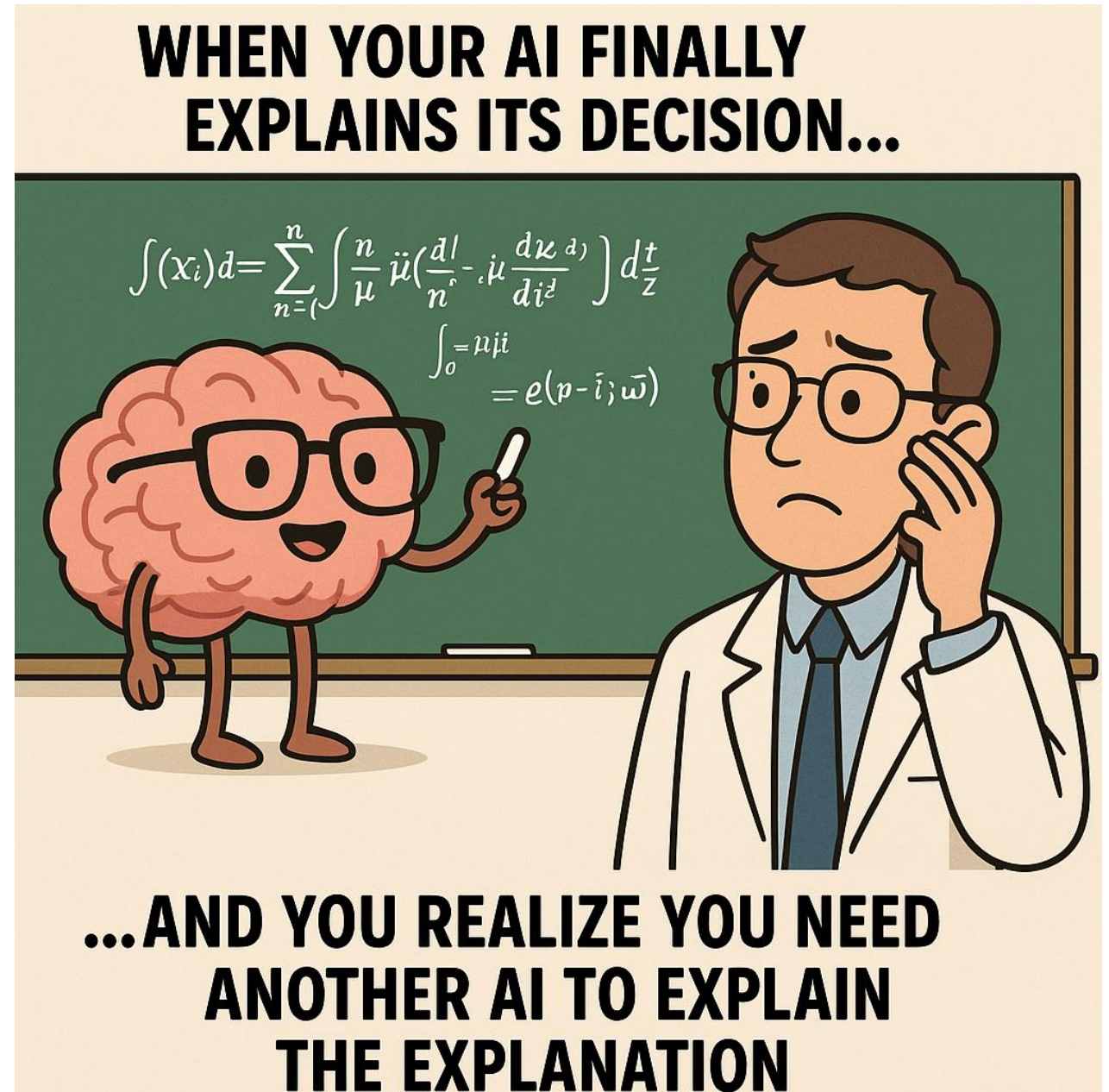
Academic Honesty

- CityU has Rules of Academic Honesty and has required all students to complete an online tutorial on subject and declare your understanding
- Plagiarism...
 - It is serious fraud to plagiarize others' work.
 - Punishment ranges from warning to course failure.
- How to prevent plagiarism...
 - Finish the assignments by yourself! You have to write the program/solution yourself.
 - okay to talk about how to do the problem with your classmates.
 - Protect your code; don't give it away as a "reference" copy.
 - In plagiarism cases, we treat the giver and the copier as both guilty.
 - You hurt your own grades by not reporting cheating.
- As instructors...
 - We have responsibility to report academic dishonesty cases so as not to compromise the quality of education.
 - We take suspected plagiarism cases very seriously.

XAI according to AI

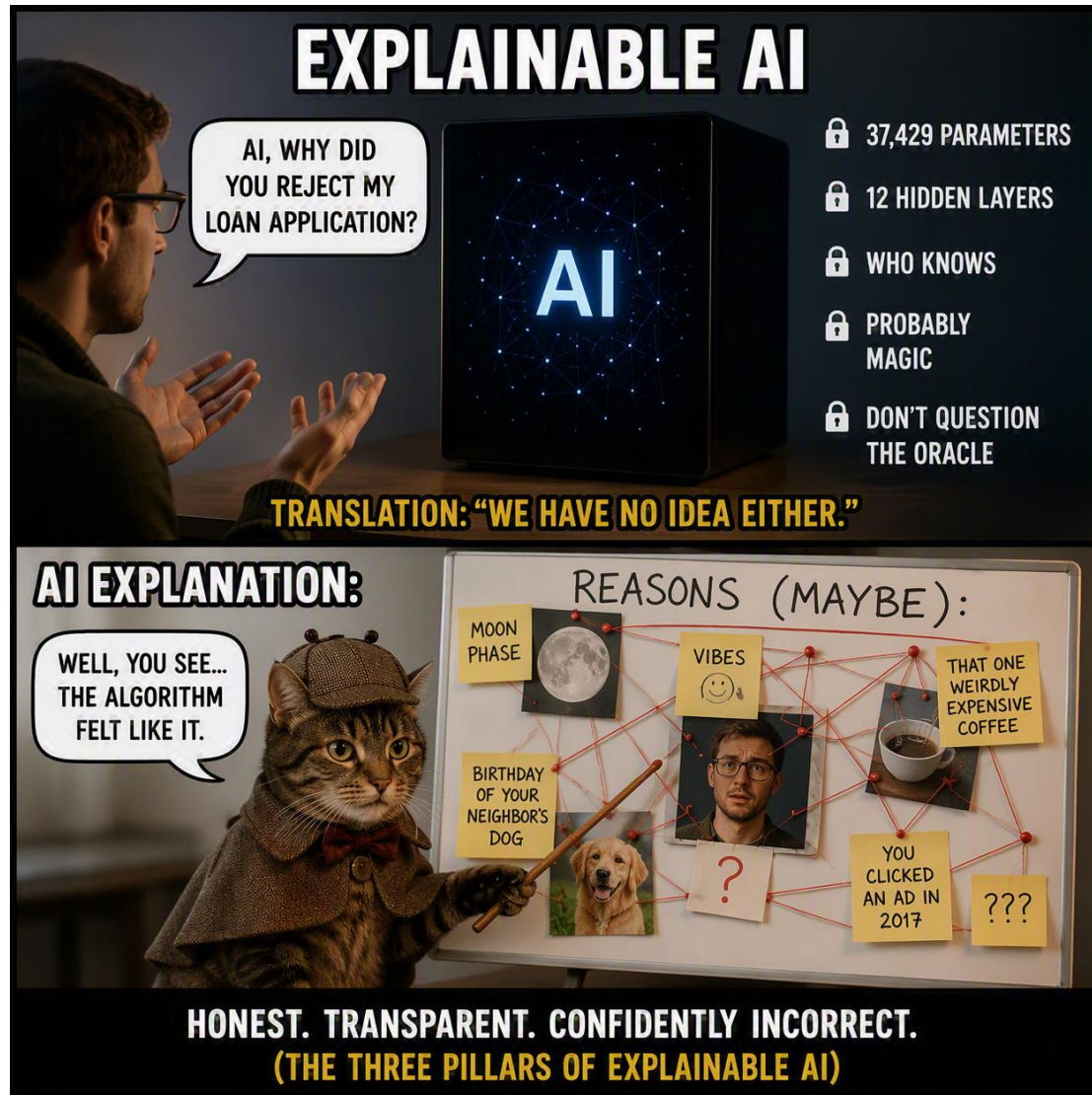


GPT-4o

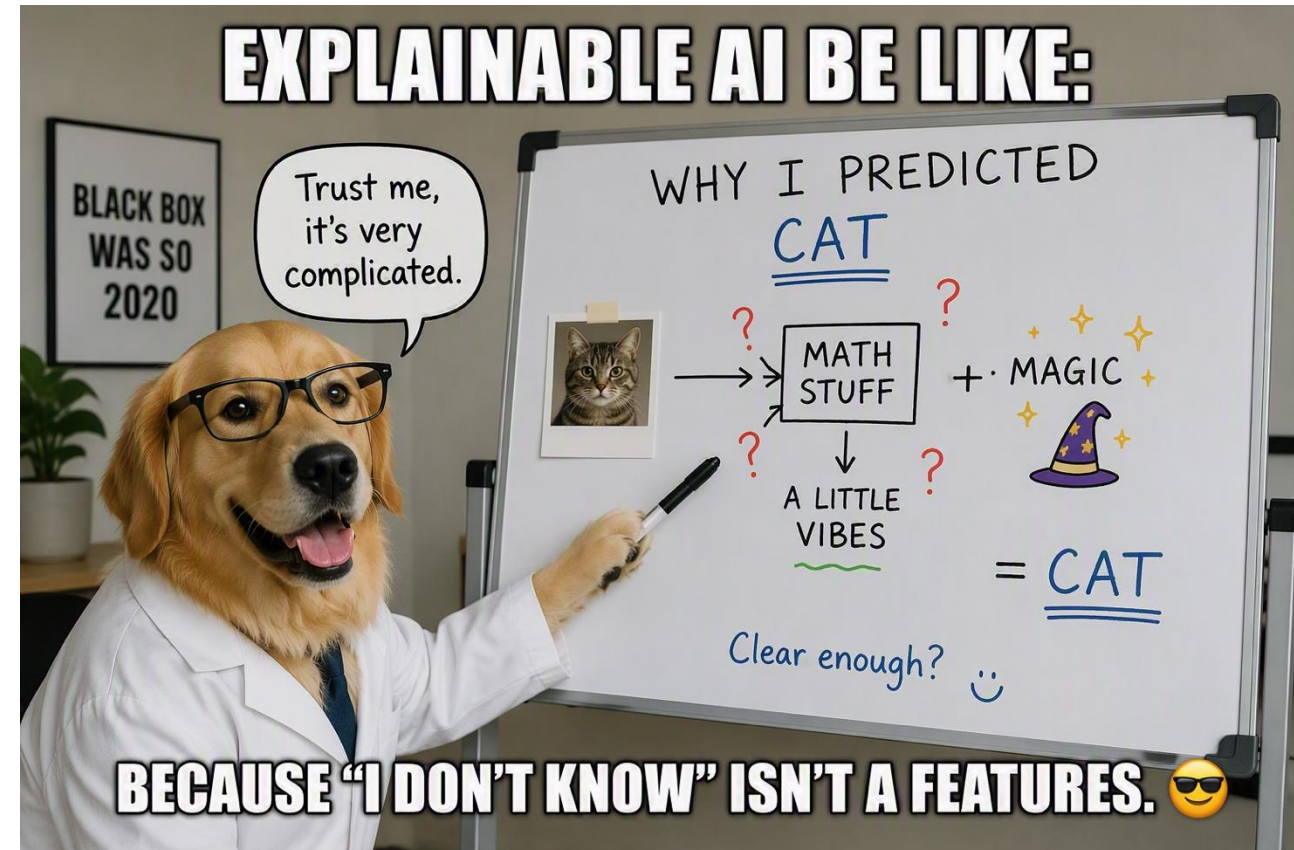


GPT-5

XAI according to AI

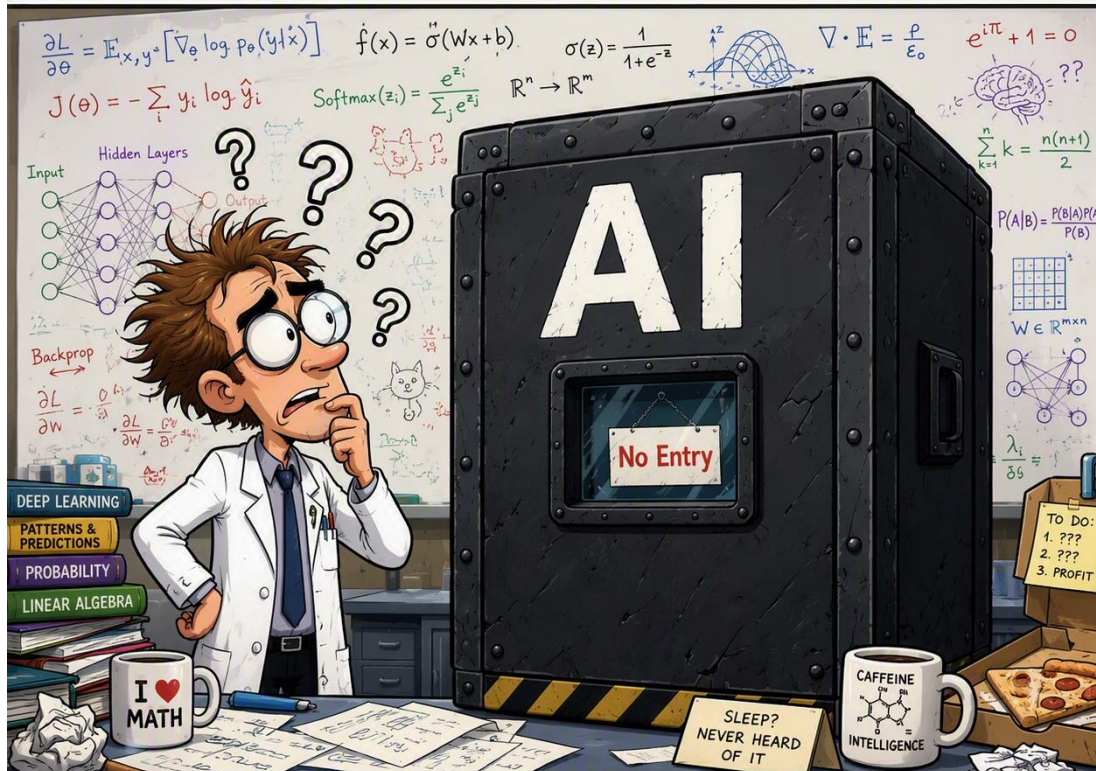


GPT-5.6 Think



GPT-5 Quick

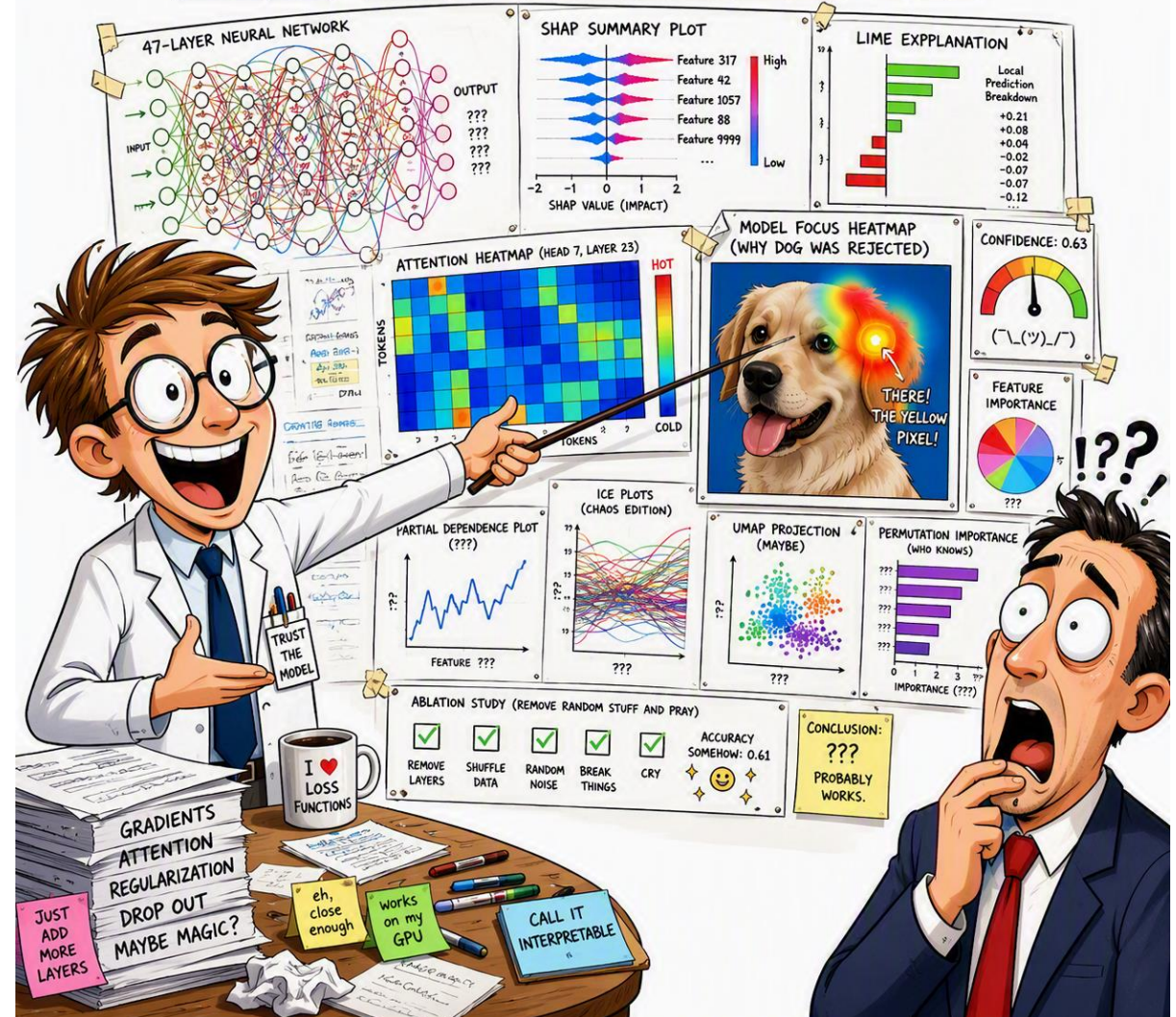
EXPLAINABLE AI RESEARCHERS:



"WE MADE IT TRANSPARENT... WE THINK."

Claude Sonnet

EXPLAINABLE AI



Claude Opus

Questions?